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PAPER

Mstnet: method for glaucoma grading based on multimodal feature fusion of spatial relations

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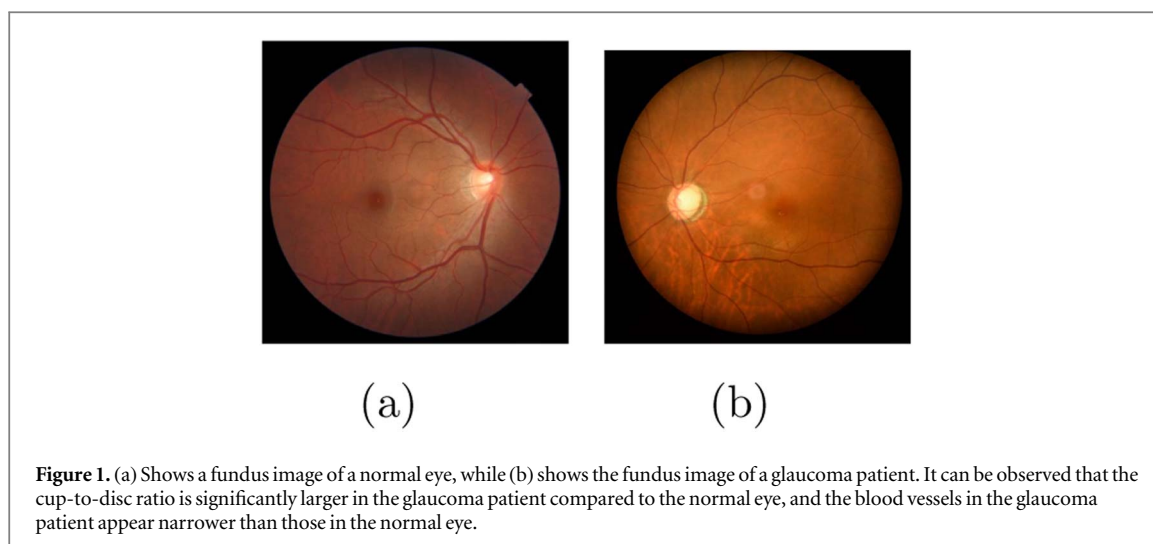
E-mail: wangxingkui@tyut.edu.cn**Keywords:** 3D transformer, glaucoma, multimodal fusion, fundus map, OCTRECEIVED
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4 December 2023**Abstract**

Objective. The objective of this study is to develop an efficient multimodal learning framework for the classification of glaucoma. Glaucoma is a group of eye diseases that can result in vision loss and blindness, often due to delayed detection and treatment. Fundus images and optical coherence tomography (OCT) images have proven valuable for the diagnosis and management of glaucoma. However, current models that combine features from both modalities often lack efficient spatial relationship modeling. *Approach.* In this study, we propose an innovative approach to address the classification of glaucoma. We focus on leveraging the features of OCT volumes and harness the capabilities of transformer models to capture long-range spatial relationships. To achieve this, we introduce a 3D transformer model to extract features from OCT volumes, enhancing the model's effectiveness. Additionally, we employ downsampling techniques to enhance model efficiency. We then utilize the spatial feature relationships between OCT volumes and fundus images to fuse the features extracted from both sources. *Main results.* Our proposed framework has yielded remarkable results, particularly in terms of glaucoma grading performance. We conducted our experiments using the GAMMA dataset, and our approach outperformed traditional feature fusion methods. By effectively modeling spatial relationships and combining OCT volume and fundus map features, our framework achieved outstanding classification results. *Significance.* This research is of significant importance in the field of glaucoma diagnosis and management. Efficient and accurate glaucoma classification is essential for timely intervention and prevention of vision loss. Our proposed approach, which integrates 3D transformer models, offers a novel way to extract and fuse features from OCT volumes and fundus images, ultimately enhancing the effectiveness of glaucoma classification. This work has the potential to contribute to improved patient care, particularly in the early detection and treatment of glaucoma, thereby reducing the risk of vision impairment and blindness.

1. Introduction: file preparation and submission

Globally, glaucoma is recognized as the second most prevalent cause of blindness. This ocular disease is a common eye disease characterized by damage to the optic nerve head and visual field defects. Glaucoma usually results from an increase in intraocular pressure, but it can also occur under normal pressure conditions. If left untreated, glaucoma can lead to blindness (Nawaldgi 2016), seriously affecting visual acuity and quality of life. The symptoms of glaucoma may not be obvious, with initial symptoms including visual field defects, elevated intraocular pressure, blurred vision, and eye pain. Due to the chronic and asymptomatic nature of early-stage glaucoma, many individuals may not seek medical attention until irreversible vision loss has occurred. As such, early detection and timely intervention are critical for preserving vision and preventing the debilitating effects of glaucoma.

Retinal fundus imaging is a non-invasive, fast, and repeatable method for examining the eyes. Retinal fundus imaging can help doctors detect and monitor various retinal diseases, such as macular degeneration, retinal tears, and retinal detachment. There are many features in the fundus image that can be used to diagnose



glaucoma, such as an enlarged optic cup (Jonas and Gründler 1996) and retinal vessel changes (Mitchell *et al* 2005). Figure 1 shows fundus images of a glaucomatous eye and a normal eye. In the caption for figure 1, the ‘disk-to-cup ratio,’ also known as the ‘cup-to-disk ratio’ or ‘C/D ratio,’ is elucidated as a spatial feature measured in fundus images. It quantifies the proportion of the optic nerve head (the optic disk) that exhibits a cup-like depression. The calculation method for this ratio involves comparing the diameter or area of the cup-like depression to the diameter or area of the entire optic nerve head.

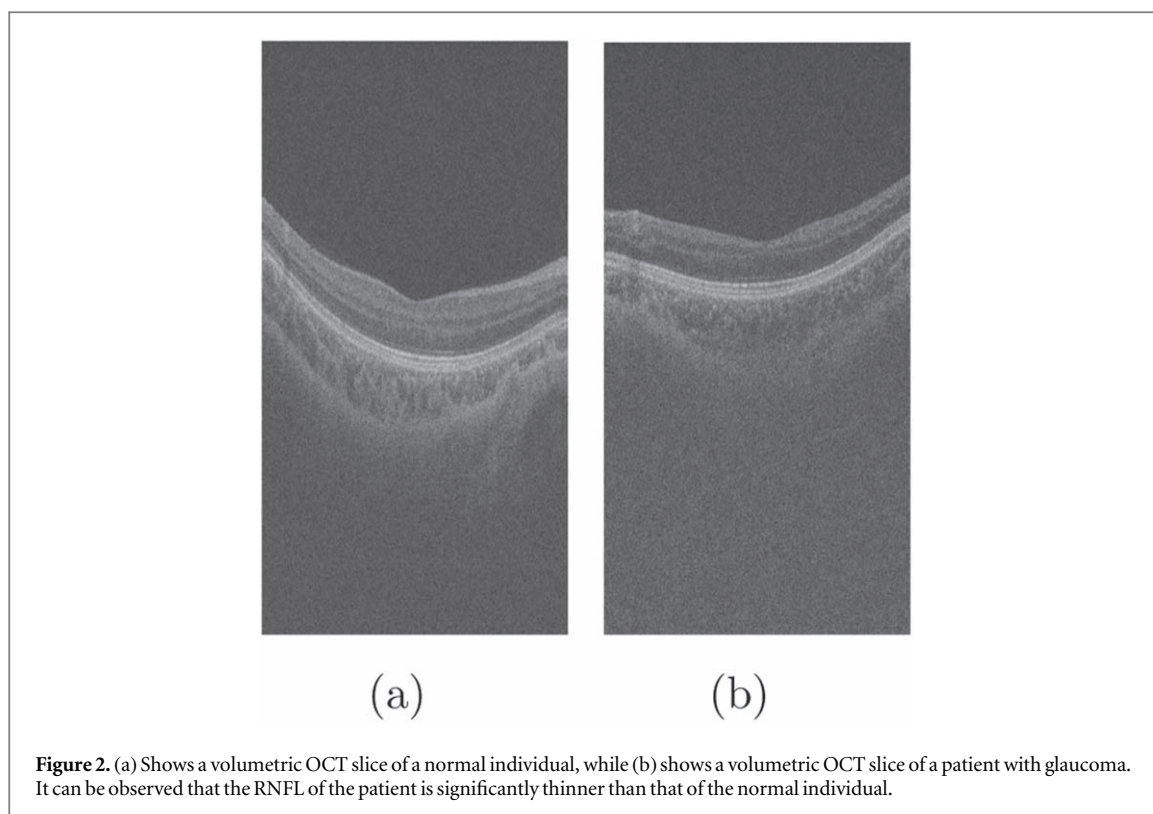
Optical coherence tomography (OCT) is a novel three-dimensional imaging technique for human tissue that enables rapid and accurate measurement of the morphology and tissue structure of various parts of the eye without contact or injury. OCT has been widely used in the diagnosis and treatment of glaucoma. The thickness and volume of the optic nerve head area (ONH) and the peripheral retinal nerve fiber layer (RNFL) in OCT images are important parameters for assessing glaucoma (Wu *et al* 2012). Within this context, ‘ONH thickness’ refers to the measurement of the thickness of the optic nerve head, which is a critical region situated at the posterior aspect of the eye, serving as the point of exit for the optic nerve. OCT imaging is capable of providing precise measurements of ONH thickness. Additionally, ‘RNFL thickness’ is a spatial feature measured in OCT images, particularly those focused on retinal imaging. It quantifies the thickness of the retinal nerve fiber layer, which encompasses the axons of ganglion cells responsible for transmitting visual information to the brain. Figure 2 shows OCT slices of a glaucoma patient and a normal person.

In recent years, some researchers have attempted to use 3D neural networks to analyze raw OCT volumes, and these studies have achieved good results (Maetschke *et al* 2019, Noury *et al* 2019, García *et al* 2021a). At the same time, transformers have also made rapid progress in the field of computer vision (Dosovitskiy *et al* 2020, Liu *et al* 2021, 2022, Touvron *et al* 2021), and transformers have demonstrated their unique advantages in the field of computer vision. Currently, some studies use both modalities and fuse the features of the two modalities (Cai *et al* 2022, Wu *et al* 2022). Compared to single modality, multimodal is often a hot research topic in medical imaging, and it can often achieve better results (Zhou *et al* 2019).

In this context, we propose an efficient multimodal fusion framework, designed for the task of glaucoma grading. We make the following contributions in our framework: as 3D transformer is more capable of extracting long-range relationships than CNN in terms of feature extraction, transformer has advantages in extracting OCT image features. Therefore, we combine Video Swin Transformer (Liu *et al* 2022) to propose a more effective method for extracting OCT image features. Furthermore, this study introduces a novel and efficient approach for multimodal fusion, which takes into account the spatial relationship between fundus images and OCT volumes. The encoded features are fused using this spatial relationship, resulting in an enhanced accuracy of the model. To validate the feasibility and effectiveness of the proposed approach, extensive experiments are conducted, yielding promising results.

Our main contributions are as follows:

- We introduced a groundbreaking innovation by transforming OCT volumes from traditional 3D images into 4D images. This pioneering approach not only preserves the essential dimension of slice stacking but also greatly enhances the subsequent spatial feature fusion between fundus images and OCT images. Furthermore, we made substantial improvements to the original feature extraction network, ensuring its seamless integration with our novel multimodal feature fusion method, thus achieving a remarkable boost in network performance and continuity.



- Our work represents a paradigm shift in the field of glaucoma research as we were the first to leverage spatial relationships for multimodal feature fusion. By incorporating a global attention mechanism, we revolutionized the way features are fused and extracted, resulting in a model that attains unprecedented levels of accuracy. This groundbreaking approach marks a significant milestone in the domain of glaucoma research.
- In a pioneering move, we conducted experiments that demonstrated the transformative potential of introducing task-specific datasets for pre-training models in cross-domain tasks. This innovative strategy led to a substantial improvement in model accuracy, emphasizing the importance of task-specific data in achieving superior results. Our research has set a new standard in the field, showcasing the power of data-driven advancements in glaucoma research.

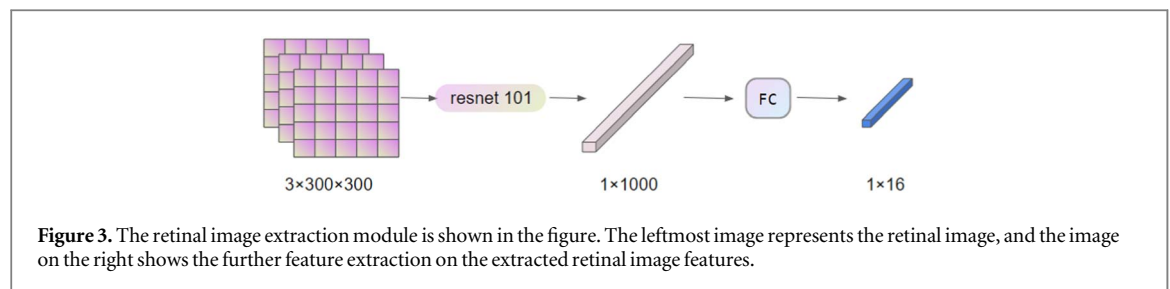
This article is structured as follows: section 2 presents a literature review of previous relevant research. Section 3 provides a detailed description of the proposed method. Section 4 offers comprehensive information on our dataset and experimental results. In section 5, we discuss our thoughts on the proposed framework and future research directions, followed by our conclusion.

2. Related work

In this section, we introduce the use of fundus images and OCT for glaucoma diagnosis, deep learning-based medical image diagnosis, and previous work on 3D image processing and related models.

Currently, many studies are based on fundus images for glaucoma diagnosis. For example, Cheng *et al* (2010) used fundus images to distinguish between closed-angle and open-angle glaucoma. Lin *et al* (2019) segmented fundus images to locate glaucoma using attention and evaluated the segmented region to diagnose glaucoma. Similarly, many studies use OCT images for glaucoma diagnosis. For example, Burgansky-Eliash *et al* (2005) effectively detected glaucoma-related changes using RNFL thickness, ONH parameters, macular parameters, and distinguished between different stages of glaucoma. Liu *et al* (2021) directly obtained RNFL thickness from spectral-domain OCT for glaucoma diagnosis, achieving good results.

The precise diagnosis of medical images using deep learning algorithms has been a long-standing issue of interest in both the medical and artificial intelligence fields. Many researchers have conducted relevant work in this area. For instance, Maetschke *et al* (2019) proposed a method that employs 3D CNNs to directly classify eyes as healthy or glaucomatous using unsegmented OCT volumes of the optic nerve head (ONH). Similarly, García *et al* (2021b) developed a CNN-based approach to grade the severity of glaucoma using circular OCT images,



achieving improved diagnostic accuracy compared to earlier methods. Another study by García-Bermúdez *et al* (2021) introduced a novel CNN-LSTM-based method that surpassed the performance of the 3D CNN approach. These studies highlight the potential of deep learning techniques for enhancing the accuracy and reliability of glaucoma diagnosis and management, and provide valuable insights into the development of advanced deep learning methods for clinical applications.

Currently, there are many studies that use fundus images or OCT images to diagnose glaucoma. In addition, many researchers have implemented glaucoma grading using multimodalities. Shigueoka *et al* (2018) used OCT and standard automated perimetry (SAP) parameters for multimodal fusion in machine learning classifiers. An *et al* (2019) and Fang *et al* (2021) developed a multimodal diagnostic algorithm based on OCT data and color fundus photographs, utilizing a deep learning model for feature extraction and a random forest classifier trained on the extracted features. In the MICCAI2021 GAMMA Challenge (Fu *et al* 2020, Orlando *et al* 2020), most participants used a feature extraction method that concatenated the encoded features from the fundus images and OCT volumes, and then used this method to achieve feature fusion for diagnosis (Wu *et al* 2022).

3D images typically contain richer information than 2D images. Compared to 2D network methods, 3D networks have the ability to extract higher dimensional features and the advantage of joint 3D spatial context information. 3D convolutional networks (Ji *et al* 2012) are an example of this, as they extract features from both the spatial and temporal dimensions through 3D convolutions to capture motion information encoded in multiple adjacent frames. In comparison to other algorithms, they have exhibited superior performance in action recognition tasks.

Compared to convolutional networks, transformers have better global feature extraction capabilities. In recent years, many excellent 3D transformer models (Bertasius *et al* 2021, Liu *et al* 2022, Tang *et al* 2022) have been proposed, and they have produced good results in various fields such as behavior recognition, 3D medical image classification and segmentation.

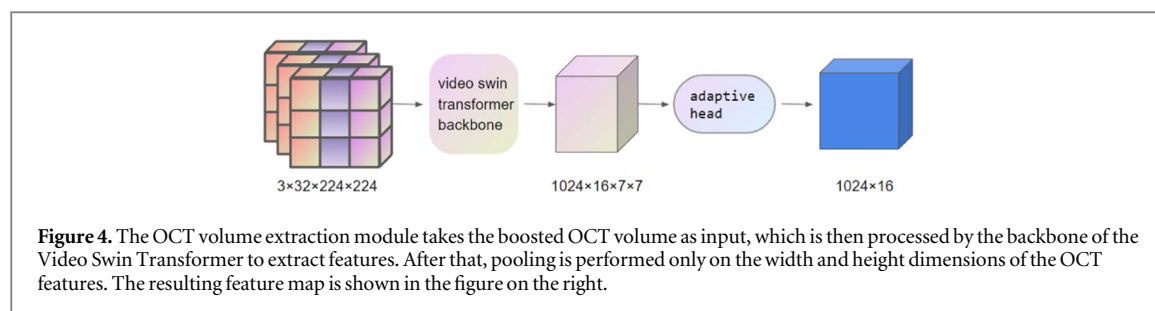
The 3D models also have their drawbacks. In terms of performance, the large parameter size of 3D models leads to high computational complexity and long computation time. Therefore, in the field of action recognition, videos are often sampled (Loshchilov and Hutter 2016, Lin *et al* 2019, Liu *et al* 2022, Tang *et al* 2022). In the medical field, many studies downsample 3D medical images (Wu *et al* 2022, Xie *et al* 2022), which can reduce the parameter size of the model and improve the training speed while maintaining a certain level of accuracy. As a data augmentation method, downsampling can effectively alleviate overfitting and improve the generalization ability of the model.

3. Methodology

In this section, we provide a detailed description of our proposed framework, which is divided into two parts. Our framework pipeline is illustrated in figure 3. In the first part, we elaborate on the feature extraction networks for both the fundus image and OCT modalities, as well as the improvements made to the original feature extraction networks. In the second part, we introduce our innovative approach for multimodal feature fusion, which leverages the spatial relationships between the features extracted from the fundus and OCT modalities. In the third part, we will describe the training method for our cross-domain model.

3.1. Feature extraction module

In this section, we will elaborate on our proposed model's feature extraction method for both fundus images and OCT volumes. We have chosen suitable networks and made improvements to their network structures to make them more adaptable to spatially-related feature fusion. The feature extraction method for fundus images and OCT volumes is as follows.



3.1.1. Extraction of fundus map features

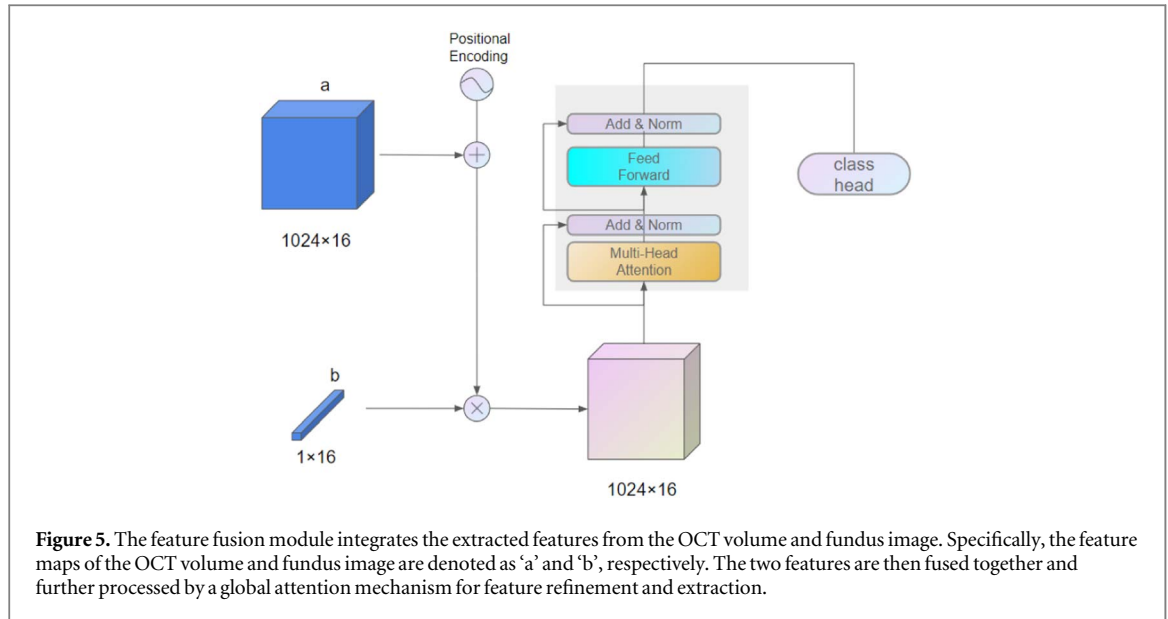
In this section, we introduce the feature extraction process for the fundus image in our model, as shown in figure 3. Based on the literature review conducted by Chen *et al* (2021), it has been concluded that the ResNet series represents the most favorable choice for extracting retinal image features in our model. This selection is substantiated by the ResNet series' exceptional performance in the realm of two-dimensional image classification, wherein it has achieved remarkable levels of accuracy. Furthermore, in comparison to models with higher accuracy, ResNet exhibits the advantages of a streamlined model architecture and a reduced parameter count. After comparing several variants through the experiments in table 6, we find that ResNet101 exhibits higher accuracy in these variants. Therefore, we chose ResNet101 as the backbone network for extracting retinal image features. Since the dimension of the feature slice after OCT feature extraction is 16, in order to facilitate multimodal feature fusion with OCT features, we further extract features from the original fundus image features. Specifically, we pass the original 1000-layer features through a fully connected layer to obtain 16-layer features.

3.1.2. Extraction of OCT features

In this section, we present an efficient method for extracting OCT volume features, as shown in figure 4. In this feature extraction method, we stack single-channel OCT slices to form 3-channel images, i.e. we increase the dimension of the $N \times H \times W$ OCT volume to $C \times N \times H \times W$ OCT volume. This approach can better preserve the N dimension features in feature extraction, facilitating the mapping of extracted features to the C dimension, which can better extract OCT volume features. Here, the C dimension represents the dimension of OCT volume channel number, and the N dimension represents the dimension of the stacked OCT slices.

Convolutional neural networks (CNNs) are good at extracting local features from images but cannot effectively establish long-range relationships between image features. However, in OCT retinal images, OCT features are often reflected in a layered manner, and the retinal nerve fiber layer (RNFL) is an important diagnostic criterion for glaucoma. This requires the model to have the ability to extract long-range relationship features, as well as to extract global or block-wise relationship features between layers. Additionally, the model needs to have the ability to extract global features between slices. The Video Swin Transformer (Wu *et al* 2022) is a pure transformer architecture for video recognition that is based on spatiotemporal localization and addresses inductive biases. This architecture has demonstrated advanced performance on datasets such as Kinetics-400, Kinetics-600, and Something v2. Compared to several well-known models with which he can achieve similar results, including UniFormerV2 (Li *et al* 2022a) and Florence (Yuan *et al* 2021), the Video Swin Transformer achieves high accuracy while utilizing only half to one-sixth of the parameters. Consequently, we have chosen the Video Swin Transformer as the backbone for feature extraction in the OCT branch of our model. Compared with 3D convolutional models, its economy in spatiotemporal modeling does not significantly compromise its performance. Compared with other networks, it fully balances the local attention and global attention in the network architecture. In this network, the T dimension (since the Video swin transformer is used for video tasks such as action recognition, the stacking dimension is located in the T -axis) features are well preserved in feature extraction, making the OCT volume features more convenient to fuse with the fundus image features using spatial relationships. We also improved the Video swin transformer as shown in figure 4, to better preserve the N -dimensional features, making it more convenient to perform multimodal feature fusion with the fundus image features in the N dimension. We improved the original I3DHead, which globally pooled over N , H , and W , to only pool over H and W , making it easier to fuse OCT volume features with fundus image features. Here, H represents the height dimension, and W represents the width dimension.

To make the model more efficient, we downsampled the OCT volume by dividing the 256 slices into 32 segments and taking one image from each segment on average. This significantly reduced the number of model parameters while maintaining accuracy, making the model more efficient.



3.2. Multimodal feature fusion of fundus map and OCT

In this section, we provide a detailed description of our spatial relation-based multimodal feature fusion method proposed in this paper.

It has been demonstrated in practice that multimodal methods can often achieve better results compared to single modality (Zhou *et al* 2019), and this has also been shown to be true in the case of glaucoma tasks (Cai *et al* 2022, Wu *et al* 2022). In the medical field, multimodal feature fusion typically emulates the diagnostic process of a doctor by concatenating encoded features for model fusion. However, this method alone may not fully achieve feature fusion.

OCT cross-sectional scans include 256 two-dimensional cross-sectional images, while fundus color photographs provide information on the tissue structure of the eye. Therefore, the N-dimensional feature vector obtained from OCT volume extraction and the feature vector extracted from fundus images can be viewed as the same dimension in space. We use the module shown in figure 5 to achieve feature fusion by leveraging the spatial relationship between the OCT volume extraction features and fundus image features. We attempt to preserve the N-dimensional features of the OCT volume and use the extracted feature vector from the fundus image as a weight, which is applied to the OCT volume feature extraction, inspired by SE-Net (Hu *et al* 2018). This approach facilitates the integration of the feature vector extracted from fundus images into the OCT volume features, thereby improving the network’s performance at minimal cost, making it more efficient. Figure 6 shows a general summary of the methodology throughout the text and is not referenced by any of the elements in the text.

In addition, we use Transformer encoder layers (Vaswani *et al* 2017) to further fuse the fundus and OCT features globally, making the information contained in the parameters of the extracted embedding richer, and the following FFN layer also enhances the model’s expression ability.

In terms of computation, the OCT extracted feature size is [B, C, N], and the fundus image feature size is [B, N]. We use X to represent the feature tensor of the OCT volume extraction and Y to represent the feature tensor of the fundus image extraction. By broadcasting the feature tensor of the fundus image to the entire fundus image globally, the operation process of the multimodal feature fusion between the OCT volume extraction feature and the fundus image extraction feature can be expressed as follows:

$$Z = X \times Y = \begin{pmatrix} x_{11}y_1 & \cdots & x_{1b}y_1 \\ \vdots & \ddots & \vdots \\ x_{a1}y_a & \cdots & x_{ab}y_a \end{pmatrix} \quad (1)$$

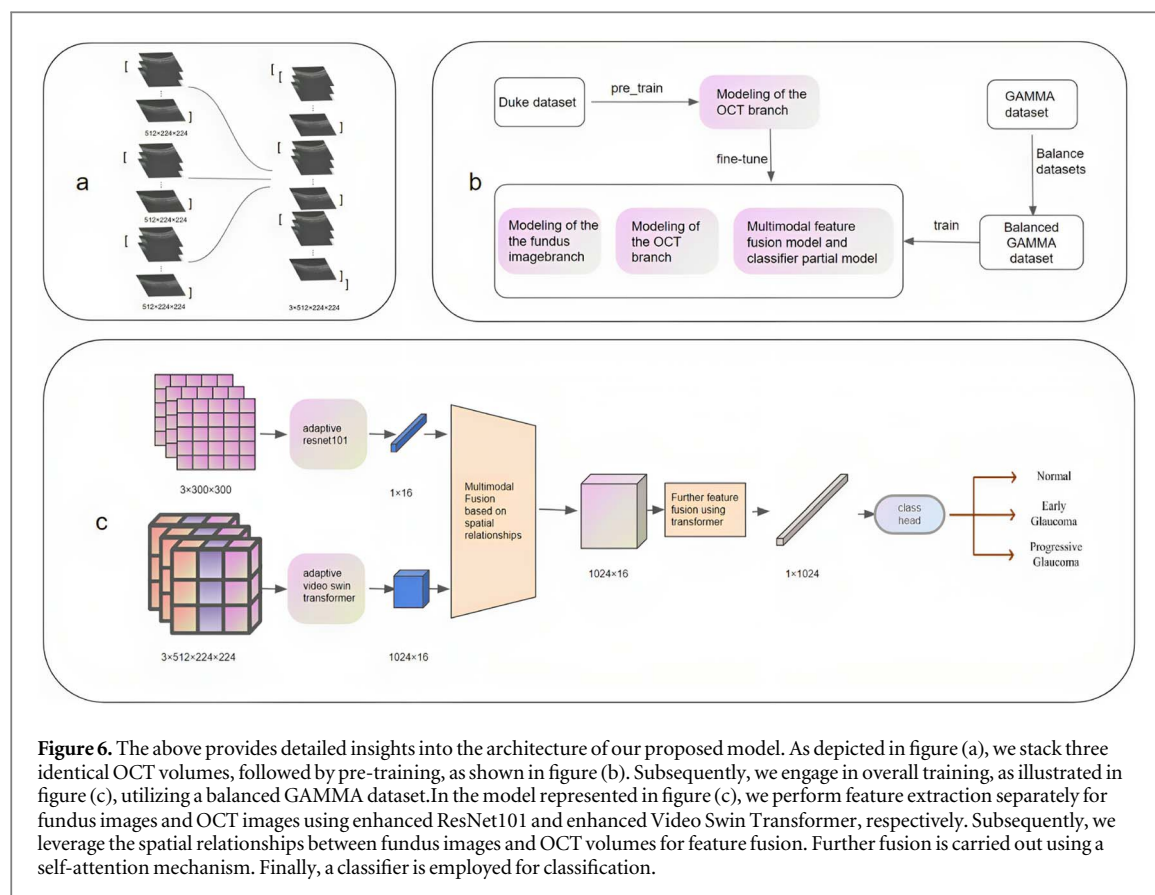
$$Z_1 = \text{Self-Attention}(Z) \quad (2)$$

$$Z_2 = \text{FFN}(Z + Z_1) \quad (3)$$

$$Z_3 = Z_1 + Z_2. \quad (4)$$

Where a denotes the dimension of N and b denotes the dimension of C.

After further multimodal feature fusion between the fundus image and OCT volume, pooling is applied to reduce the dimensionality of the features. The final output is generated by the classification head, which not only performs further feature extraction but also produces the final result.



3.3. Loss function

During the training stage, we used the cross-entropy loss function, which is expressed in formula as follows:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^C y_{i,j} \log(p_{i,j}) + (1 - y_{i,j}) \log(1 - p_{i,j}),$$

where N is the number of training samples, C is the number of classes, and the value of C in this paper is 3, $y_{i,j}$ is the ground truth label of the i -th sample for class j , and $p_{i,j}$ is the predicted probability of the i -th sample for class j . The loss function penalizes the model when the predicted probability is far from the ground truth, and guides the model to minimize the overall error in the classification task. During the training process, the weights of the network are updated by backpropagation to minimize the loss function.

3.4. Training methods for cross-domain models

In this section, we describe how we apply pre-trained models in the action recognition domain to medical tasks with limited data, as encountered in this paper. Due to the large number of parameters in the Video Swin Transformer, it is relatively difficult to train compared to other models. Therefore, we use a pre-trained model trained on Kinetics-400. However, when using this pre-trained model, the performance indicators of the framework are not very good. To address this, we introduce a publicly available dataset from Duke University (Farsiu *et al* 2014), which was developed in collaboration with Farsiu *et al* from the Devers Eye Institute, Duke Eye Center, Emory Eye Center, and the National Eye Institute (Farsiu *et al* 2014). This dataset contains a large number of OCT volumes of the eye, which is more similar to our task than the GAMMA dataset used in this paper. We trained the original pre-trained model on this dataset, resulting in a second round of training. Compared to using the pre-trained model directly, the framework trained with the second round of training achieved significant improvements in model accuracy and other performance metrics, as well as enhancing the model's interpretability.

4. Experiments and results

In this section, we use two datasets to train our proposed model: a publicly available dataset from Duke University and the GAMMA dataset (Wu *et al* 2022). We will first introduce each dataset in detail. Then, we will

Table 1. The number of OCT volumes in the part of the DUKE dataset used in this study.

Grade	Category	Number of samples
0	AMD	269
1	Control	115
sum	—	384

Table 2. Number of samples in the GAMMA dataset.

Grade	Categories	Number of samples
0	No glaucoma	50
1	Early glaucoma	26
2	Moderate or advanced glaucoma	24
Sum	—	100

describe the experimental process in detail. Finally, we will provide comparison results with other algorithms and visualization results to evaluate our algorithm.

4.1. Dataset

We trained our model using the Duke University open database and the GAMMA dataset. Since the pre-trained model we used, Video swin transformer, was trained on the Kinetics-400 dataset, it could not be used as a good pre-trained model for glaucoma grading training. Therefore, we used the model obtained by fine-tuning the pre-trained model using the Duke University open database as the OCT branch of our multi-modal model. Subsequently, we trained the multi-modal model using the GAMMA dataset and compared it with other models.

4.1.1. Introduction to the Duke dataset

The Duke OCT Retinal Data Repository is a publicly available database that contains a large number of retinal OCT images and related clinical information. In this project, we used the dataset collaboratively researched by Farsiu *et al* with the Devers Eye Institute, Duke Eye Center, Emory Eye Center, and the National Eye Institute in the United States (Farsiu *et al* 2014). In the dataset, 38,400 OCT images were collected and made publicly available, each consisting of 1000 A-scans. The subjects were AMD patients and healthy volunteers aged between 50 and 85 years, and the AMD patients had not undergone vitreoretinal surgery or had any ophthalmic diseases affecting vision. Their eyes were either in the intermediate stage of AMD or one eye was in the intermediate stage and the other in the late stage. Table 1 shows the distribution of the DUKE dataset used in this study.

4.1.2. Introduction to the GAMMA dataset

The GAMMA dataset (Wu *et al* 2022) released by Guangzhou Zhongshan Ophthalmic Center of Sun Yat-sen University, China, consists of 300 retinal OCT pairs corresponding to Chinese patients aged 19–77 years with a mean age of 40.6414.53 years. The dataset includes samples from four categories (non-glaucoma, early glaucoma, moderate glaucoma, and advanced glaucoma), with each category randomly divided into three roughly equal parts corresponding to the three tasks of the GAMMA Challenge (multi-modal glaucoma grading, macular fovea localization, and optic cup and disc segmentation). In this paper, we use the multi-modal glaucoma grading dataset consisting of 100 pairs of retinal and OCT images from the GAMMA dataset, which have been labeled by ophthalmologists as one of the three levels of glaucoma severity ('none,' 'early,' and 'moderate-to-severe'). Each retinal image is either 2000×2992 pixels or 1934×1956 pixels in size, and each OCT volume consists of 256 2D slices of size 992×512 pixels. Table 2 shows the distribution of the GAMMA dataset used in this paper. In addressing the issue of sample imbalance within the GAMMA dataset, we employed the RandomOverSampler technique. RandomOverSampler represents a machine learning approach specifically designed for mitigating the challenges posed by imbalanced datasets. Its primary objective lies in rectifying the disparate distribution of samples among different categories by generating supplementary instances. This is accomplished through the random selection of a subset of samples from the minority class, subsequently replicating them multiple times. This process serves to facilitate more effective learning of the distinctive features associated with the minority class, ultimately leading to an enhancement in overall model performance.

4.2. Implementation details

The implementation details of the experimental and testing phases are presented as follows.

4.2.1. Experimental and implementation details

All of our experiments were implemented using PaddlePaddle 2.2.0 and Tesla V100. In the first stage of training, we used a pre-trained Video Swin Transformer model that was trained on the Kinetics-400 dataset. However, as this dataset focuses on human-object and human-human interactions in video clips, it did not perform well for medical tasks. Therefore, we chose the DUKE dataset, which has a larger volume of data compared to the GAMMA dataset, to retrain the OCT branch. We divided each OCT volume into 32 slices, and took one slice from each division. We then stacked these three single-channel OCT slices to form a three-channel OCT volume, which transformed the dimensions of the OCT volume from three-dimensional to four-dimensional. We then centered and resized the image to 512×512 pixels and 224×224 pixels, respectively, to match the expected resolution of the Video Swin Transformer. We set the batch size to 4, used SGD as the optimizer, and applied linear learning rate warm-up for the first 300 iterations, with a learning rate of $2e-4$. After that, we applied cosine decay (Loshchilov and Hutter 2016) and used cross-entropy loss (Lce) as the loss function.

In the second stage of training, we addressed the data imbalance and scarcity issues in the GAMMA dataset by resampling the ‘early’ and ‘intermediate-high’ parts of the dataset to achieve balanced data. We then preprocessed the images as follows: for fundus images, we resized them to 300×300 pixels; for OCT volumes, we transformed all slices into three-channel images and read 32 slices from each OCT volume slice, using the same method as in the first stage of training. We then centered and resized the image to 512×512 pixels to remove the black borders and 224×224 pixels, respectively. For the fundus image feature extraction branch, we used the ResNet101 model pre-trained on ImageNet and modified it as described in section 3.1.1 for feature extraction. For the OCT volume feature extraction branch, we removed the classification head from the model generated in the first stage of training and modified it as described in section 3.1.2 for feature extraction. We then used the feature fusion method described in section 3.2 to fuse the extracted features. We set the batch size to 4, used SGD as the optimizer, and applied linear learning rate warm-up for the first 300 iterations, with a learning rate of $1e-4$. After that, we applied cosine decay (Loshchilov and Hutter 2016) and used cross-entropy loss (Lce) as the loss function. To make more effective use of the data, we performed two rounds of random splitting and two rounds of training to improve the kappa coefficient of the test results.

To enhance the diversity and generalizability of our dataset, we employed various data augmentation techniques during model training. Specifically, for the fundus image branch, we applied random center cropping with a range of $[0.8, 1.2]$, random horizontal and vertical flipping, as well as random rotation by 180 degrees. In the case of the OCT volume branch, we utilized random horizontal flipping and rotation by 180 degrees. These augmentation methods were chosen based on their ability to introduce realistic variations to the training data and reduce overfitting. By incorporating these techniques, we aimed to improve the robustness of our model and enhance its ability to accurately classify glaucoma severity levels.

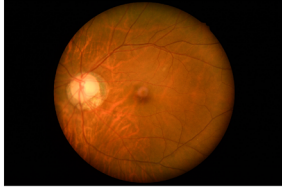
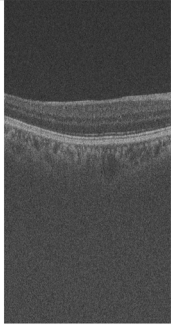
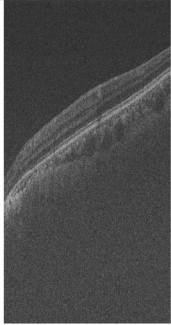
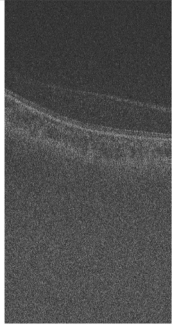
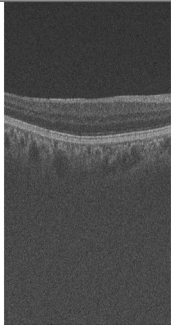
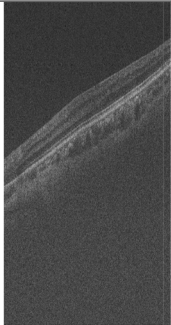
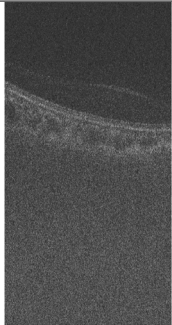
4.2.2. Evaluation metric

In this study, we employed a quantitative evaluation metric, Cohen’s kappa coefficient (kappa) (McHugh 2012), as the evaluation measure for the three-class classification problem. The kappa coefficient in this paper was calculated based on the confusion matrix. Kappa is a statistical measure used to assess the accuracy of a classification model, which is calculated by comparing the consistency between the predicted results of the classification model and the actual observed results. The range of kappa coefficient is between -1 and 1, where 0 indicates that the predicted results of the classification model are not related to the actual observed results, 1 indicates that the predicted results of the classification model are completely consistent with the actual observed results, and -1 indicates that the predicted results of the classification model are completely opposite to the actual observed results. Its formula can be expressed as:

$$K = \frac{p_o - p_e}{1 - p_e}, \quad (5)$$

where p_o is the observed agreement proportion and p_e is the expected agreement proportion by chance. Specifically, for c categories, let a_{ij} denote the number of subjects predicted as j and labeled as i . Let the total number of subjects be $n = \sum_{i=1}^c \sum_{j=1}^c a_{ij}$. Then:

Table 3. In this table, we have extracted one sample each from the detection results of ‘no glaucoma,’ ‘early glaucoma,’ and ‘moderate or advanced glaucoma’ categories. We present the fundus images from each of these samples. In addition, we selected the 85th and 171st images from the 256 OCT images per sample that followed the systematic sampling method and displayed them.

	non	early	mid advanced
fundus image			
85th OCT image			
171st OCT image			

$$p_o = \sum_{i=1}^c \sum_{j=1}^c w_{ij} \frac{a_{ij}}{n}$$

$$p_e = \sum_{i=1}^c \sum_{j=1}^c w_{ij} \frac{(\sum_{k=1}^c a_{ik} \times \sum_{k=1}^c a_{kj})}{n^2}.$$

In our evaluation, w_{ij} is the weight assigned to a correct prediction, the weight assigned to a prediction with one level of deviation is 0.5, and the weight assigned to a prediction with two levels of deviation is 0.

Given that our sample is unevenly distributed across each category, we use Kappa as it is a good evaluation metric for imbalanced datasets.

4.3. Result

In this section, we present and analyze the results of our experiments, as illustrated in table 3. Additionally, we compare our approach with those of other researchers.

To assess the efficacy of our proposed model, we conducted a comprehensive evaluation of its performance through a series of experiments and subsequent analysis of the results. Three groups of experiments were carried out, as presented in table 4. The first group involved a comparison between the results obtained by utilizing our proposed multimodal fusion approach, which combines fundus and 3D OCT images, and the results produced by using only the fundus image branch or OCT volume branch, trained using the methodology proposed in this

Table 4. We present in this paper the results of the ablation analysis of the model for glaucoma grading. OCH indicates that our proposed multimodal feature fusion scheme is used.

Modeling of Fundus branching	Color fundus photography	3D OCT	OCH	Kappa ↑
resnet101	✓			0.673
		✓		0.650
resnet101	✓	✓		0.759
resnet50	✓	✓	✓	0.860
resnet152	✓	✓	✓	0.835
resnet101	✓	✓	✓	0.892

paper. Notably, the kappa coefficients obtained from using only the fundus image branch or OCT volume branch were 0.673 and 0.650, respectively. However, the kappa coefficient obtained from our proposed multimodal fusion approach was significantly higher. These results highlight the superior performance of our proposed model and its ability to effectively integrate information from both fundus and OCT images, enabling more accurate classification of glaucoma severity levels. The findings of this study provide valuable insights into the development of robust and efficient deep learning-based models for clinical decision support in glaucoma diagnosis and management. In the second set of experiments, we compared the performance of our proposed model with traditional approaches. The traditional approaches encompassed a comprehensive procedure for feature extraction, fusion, and classification. Initially, distinctive features were independently extracted from both the fundus images and OCT images. Subsequently, the extracted features from the two distinct modalities, fundus and OCT, were fused together through a concatenation operation. This operation formed a consolidated feature vector, effectively integrating the complementary information originating from the two imaging sources. The feature concatenation step was a pivotal stage, aimed at creating a single representation that encapsulated the salient characteristics from both fundus and OCT data. In the classification phase, a series of fully connected layers, also known as dense layers, were employed as classifiers within the neural network architecture. These layers played a crucial role in mapping the concatenated feature vector into the respective target classification categories, enabling the categorization of fundus and OCT data into their designated classes or diagnostic categories based on the fused features. The experimental results showed that our proposed multi-modal feature extraction method outperformed the traditional method, with a kappa coefficient of 0.892 compared to 0.759. In the third set of experiments, we modified only the fundus image branch by replacing it with various other variants of the ResNet architecture. These modifications were carried out to facilitate a comparative analysis with our original model. The quantitative results obtained from our experiments provide compelling evidence supporting the effectiveness of our proposed model in enhancing the accuracy of glaucoma classification.

Furthermore, we compared the results of using a pre-trained model directly from the Kinetics-400 dataset with the results of using a model that was fine-tuned with the Duke University open database. As shown in table 5, the model that was fine-tuned with the task-specific dataset significantly outperformed the model trained directly on the pre-trained Kinetics-400 dataset, with a kappa coefficient of 0.711. This result highlights the importance of fine-tuning pre-trained models for medical tasks with a limited number of samples and in a different domain.

In our feature fusion module, as compared to schemes that perform feature fusion during feature extraction, such as the approach proposed by Li *et al* (2022b), our method evidently enjoys a natural advantage. This advantage arises from our approach of separately invoking pre-trained models for each branch. We conducted a comparative analysis between our approach, which fuses features after feature extraction, and similar methods that perform feature fusion following a similar procedure. In table 6, we conducted a comparative analysis by introducing the multimodal fusion components from other well-established networks in the field of ophthalmology into our model, while retaining the backbone of our network. Yoo *et al* (2019) employed a random forest approach to achieve feature fusion between fundus images and OCT scans. Yang *et al* (2021) employed an averaging fusion strategy to implement a bimodal deep learning approach for the recognition of various retinal diseases based on color fundus photographs and OCT images. Xiong *et al* (2022), in their work on detecting glaucomatous optic neuropathy through the use of visual field (VF) and OCT data, proposed the incorporation of an attention module. This module leveraged a weighted averaging technique to obtain a comprehensive representation from the bimodal features, thereby achieving multimodal feature fusion.

In table 7, we compare our model with state-of-the-art (SOTA) methods for glaucoma grading. García *et al* (2021b) and Cheng *et al* (2011) evaluated their proposed models on other glaucoma datasets with over 1000 samples. SmartDSP team (Maetschke *et al* 2019) evaluated their model on the MICCAI2021 Contest: GAMMA

Table 5. We compared the results obtained from fine-tuning the pre-trained model with those obtained by directly using the pre-trained model. The results are presented below.

	Kappa ↑
The use of fine-tuning	0.711
The absence of fine-tuning	0.892

Table 6. Comparing our multimodal feature fusion component with the multimodal feature fusion component of state-of-the-art networks.

Methods for fusing multimodal features	Kappa ↑	acc ↑	F1-score ↑
Yoo <i>et al</i> (2019)	0.696	0.74	0.66
Yang <i>et al</i> (2021)	0.657	0.72	0.65
Traditional method	0.759	0.72	0.63
Xiong <i>et al</i> (2022)	0.825	0.77	0.69
Mstnet	0.892	0.91	0.88

Table 7. Comparison of our model and SOTA methods on glaucoma grading task.

Methods	Dataset	Kappa ↑	acc ↑
García <i>et al</i> (2021b)	Private	0.719	0.818
Cheng <i>et al</i> (2011)	Private (Cheng <i>et al</i> 2010)	0.641	0.863
Corolla (Cai <i>et al</i> 2022)	GAMMA	0.855	0.798
SmartDSP (Wu <i>et al</i> 2022)	GAMMA	0.855	0.900
Mstnet	GAMMA	0.892	0.910

dataset. Our proposed model performs much better in terms of Kappa than all four comparison methods. It is important to acknowledge that a fair comparison of the five sets of results may not be feasible due to the varying datasets on which they are based. Notably, the GAMMA dataset, being the most recent release, has limited reported results available for comparison against our findings. As such, we endeavored to evaluate our results in relation to those derived from relevant tasks in the literature. This approach allowed us to draw meaningful comparisons and provide context for the efficacy of our methodology. By leveraging existing benchmarks and performance metrics, we were able to establish a clear understanding of the strengths and limitations of our approach, and identify areas for further improvement.

Results show that our feature fusion method is much better than other multimodal fusion methods. The use of spatial relations for feature fusion clearly results in a more comprehensive fusion than other methods of multimodal feature fusion.

5. Discussion

The above experiments demonstrate the effectiveness of our algorithm. In medical image processing, annotations usually require personnel with years of professional training to perform, which is costly and difficult to obtain. Therefore, it is challenging to collect large datasets. Consequently, it is almost impossible to use medical data for pre-training image models. In comparison to medical data, using traditional data to pre-train models is much more convenient. For example, training the Video Swin Transformer with medical data is almost impossible. Therefore, using Kinetics-400 as the dataset would significantly reduce the cost of training the Video Swin Transformer model. However, such pre-trained models are not conducive to training medical models. On the one hand, this pre-training model will reduce the accuracy of the medical model, and on the other hand, it will make the interpretability of the medical model worse. Therefore, we need to introduce relevant medical datasets to train the model again.

This paper demonstrates the superiority of using spatial relationships for feature fusion. At the same time, the features generated by fusing features with spatial fusion can be further extracted or fused by the network, making the feature fusion more sufficient. Therefore, using spatial features to achieve multimodal fusion will

greatly improve the fusion degree of multimodal features, and the features obtained from the final classification evaluation are more fully fused features, greatly improving the accuracy of the model.

The multimodal fusion framework proposed in this paper aims to achieve more comprehensive feature fusion of multimodal features using spatial relationships, and can also be extended to other diseases, such as macular degeneration and diabetic retinopathy (DR). Whether our framework also helps to classify these diseases is one of our future work. At the same time, our framework can also be extended to the feature fusion of other 2D+3D images, such as breast magnetic resonance imaging and three-dimensional breast x-ray photography to check breast conditions, or using chest radiographs and CT images to check lung conditions. Whether our framework is applicable to these images is also our future work. These works can make our network more general.

The feature fusion method proposed in this paper is not only applicable to the fusion of spatial features but can also replace the part of the network that extracts features for extracting other features. For example, replacing the feature extraction part with a network that extracts temporal features to examine the patient's heart condition can be used to extract electrocardiogram features. Using an image feature extraction network to extract ultrasound cardiogram features can fuse temporal and spatial features. In future work, we will consider replacing the feature extraction network in the framework to apply the multimodal feature fusion method proposed in this paper to other types of multimodal feature fusion, making our multimodal feature fusion method more universal.

6. Conclusion

In this work, we propose and validate an efficient multimodal fusion framework. In this framework, we successfully introduce a new method for extracting OCT volumes, making the extraction of OCT volume features more efficient. Additionally, we observe that using spatial relationships for multimodal feature fusion is more effective than traditional multimodal feature fusion methods in glaucoma grading tasks. However, it should be noted that the size of the GAMMA dataset used in this work is relatively limited, and thus, further exploration is required to investigate the generalizability of our framework to other tasks. Future research will be necessary to assess the applicability and effectiveness of our proposed methodology in diverse settings.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary information files).

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